GENERATIVE LARGE LANGUAGE MODEL-POWERED CONVERSATIONAL AI APP FOR PERSONALIZED RISK ASSESSMENT: CASE STUDY IN COVID-19

by

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THESIS

Submitted to the Graduate School,

of Wayne State University,

Detroit, Michigan

in partial fulfillment of the requirements

for the degree of

MASTER OF SCIENCE

2025

MAJOR: COMPUTER SCIENCE

Approved By:

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ACKNOWLEDGEMENTS

I would like to express my deepest gratitude to my advisor, Dr. Dongxiao Zhu, for his guidance, support, and encouragement throughout the development of this work. I am especially thankful to Xiangyu Zhou, whose insights, technical help, and many discussions significantly strengthened the design of the application and the experimental framework. I also appreciate Yao Qiang for his valuable suggestions on the experiments and manuscript revisions.

My sincere thanks go to Srinivasan Suresh, Steven Hicks, and Usha Sethuraman for their contributions to data collection and for the helpful feedback that shaped several stages of this project. I am grateful to the members of the Trustworthy AI Lab for creating an environment that supported both the research and implementation aspects of this work.

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CHAPTER 1 INTRODUCTION

1.1 Background

Disease risk assessment is a critical component of public health surveillance, where demographic variables and social determinants of health are used to estimate a patient's susceptibility to disease, predict treatment response, and forecast severity outcomes [20, 31, 21]. Traditionally, these predictions rely on machine learning models trained de novo for each disease using curated structured (tabular) data. For example, Earlier work by Wang et al. demonstrated how multi-level risk factor modeling can be applied to clinical risk prediction using deep neural architectures [31], while Li et al. proposed a deep mixture neural network to stratify patients and predict heart failure risk within each subgroup [21]. These approaches can achieve strong performance but depend heavily on feature engineering, labeled tabular data sets, and bespoke model development for each condition.

The advent of transformer architectures has shifted this paradigm by enabling models that can capture complex dependencies in clinical data with minimal manual feature design. Bidirectional Encoder Representations from Transformers (BERT) introduced a pretraining–finetuning recipe that has been widely adopted in health care applications [10]. Domain-specialized variants such as ClinicalBERT and BioClinicalBERT, trained on MIMIC-III clinical notes, improve performance on tasks like mortality prediction and readmission risk [16, 3]. Med-BERT extends this idea to large-scale structured electronic health records (EHRs) and achieves strong area under the curve (AUC) values for disease prediction across multiple phenotypes [25]. Despite their success, these BERT-style models are primarily discriminative encoders optimized for classification or regression and are less

naturally suited to interactive, streaming question–answer (QA) workflows that characterize conversational risk assessment tools.

In many real-world settings, risk assessment is not a one-shot prediction from a static feature vector but a dynamic process: patients provide partial information, ask clarifying questions, and update their responses over time. Conventional discriminative models struggle to ingest such mixed, evolving inputs, especially when they combine categorical responses, free text, and contextual metadata. This motivates moving from task-specific discriminative models toward more flexible, generative architectures that can both understand and produce natural language while grounding their predictions in clinical variables (Figure 1).

1.2 Generative Large Language Models for Health Care

Large, generative language models such as GPT-3 have overcome several limitations of discriminative encoders by handling heterogeneous data formats and supporting natural language interaction [7]. These models can jointly process structured clinical attributes and unstructured narratives, integrating information from patient histories, symptom descriptions, and contextual factors to support complex tasks such as disease severity prediction and triage. As a result, generative LLMs have been explored for diagnostic support, clinical decision-making, knowledge extraction, and personalized risk monitoring across a broad range of conditions.

In diagnostic support, models like ChatGPT and GPT-4 have been evaluated on standardized medical examinations and simulated clinical cases. Gilson et al. assessed ChatGPT on USMLE Step 1 and Step 2 multiple-choice questions and showed that its performance

approached or exceeded the passing threshold, highlighting its potential in medical education and diagnostic reasoning [11]. Kung et al. tested ChatGPT on USMLE-style clinical vignettes to examine its capacity for clinical reasoning without medical-specific finetuning [19]. Beyond exam-style questions, Ali et al. used ChatGPT to generate patient-friendly clinical letters from semistructured physician inputs, demonstrating how LLMs can support communication and documentation [2]. Xu et al. applied ChatGPT to urological diseases using semistructured patient data and found that it can produce plausible differential diagnoses, suggesting a role as an assistive tool in preliminary diagnostic workups [33]. Kanjee et al. evaluated GPT-4 on a complex diagnostic challenge and showed that it can generate reasonable diagnostic hypotheses, though with important limitations that necessitate clinician oversight [17]. Complementary to these targeted evaluations, Nori et al. characterized GPT-4's capabilities across a range of medical challenge problems, providing a broader view of its strengths and failure modes in clinical reasoning [24].

Generative LLMs are also increasingly used as tools for synthesizing medical evidence. Tang et al. evaluated large language models on summarizing medical literature and showed that models like GPT-4 can generate concise, clinically relevant summaries of research articles, potentially reducing the burden of staying up to date with emerging findings [29]. Sallam surveyed the role of LLMs in systematic reviews and meta-analyses, outlining how they can assist in literature search, screening, and data extraction while emphasizing the need for human verification [26]. These studies suggest that generative LLMs can act as "literature co-pilots," enabling clinicians and researchers to navigate rapidly expanding evidence bases more efficiently.

Beyond knowledge and diagnostic tasks, several recent works directly use generative

LLMs for risk prediction and personalized monitoring. Health-LLM integrates conversational inputs and wearable sensor data, such as physical activity and heart rate, to predict stress, fatigue, and related health metrics in a zero-shot manner [18]. Instead of training a new model for each outcome, Health-LLM leverages the generalization capabilities of a pre-trained LLM to adapt to multiple prediction tasks with minimal labeled data. ClinicalMamba extends this line of research by introducing a generative architecture designed specifically for longitudinal clinical notes, enabling improved modeling of temporal clinical patterns [34]. Together, these efforts illustrate a broader shift toward using generative LLMs as unified engines that can handle both natural language and structured signals for a variety of clinical prediction problems.

1.3 Foundation Models Specialized for Medicine

While general-purpose LLMs such as GPT-3 and GPT-4 exhibit strong zero-shot performance, several groups have developed foundation models specialized for medicine. Med-PaLM and Med-PaLM 2 fine-tune large language models on curated medical QA data sets and demonstrate near expert-level performance on benchmark question-answering tasks, including those derived from standardized exams and real-world clinical queries [28]. PMC LLaMA provides another example of domain-specialized pretraining, using PubMed Central articles to adapt the LLaMA architecture for medical reasoning tasks [32].

Beyond pure text, foundation models for medicine increasingly combine structured and unstructured data. CPLLM, for example, explores clinical prediction with large language models by conditioning on structured EHR-style inputs and generating outcome probabilities through language modeling objectives [4]. Llama 2, a family of open-weight foundation

models, has been widely adopted as a base for medical finetuning due to its scalability and permissive licensing [30]. BioMedLM is another open biomedical language model trained on large biomedical corpora and positioned as a base for downstream medical NLP tasks [5]. These specialized models provide an ecosystem of architectures that can be adapted to disease risk prediction, prognosis, and patient-facing applications.

A parallel line of work investigates how LLMs handle tabular or mixed-modality inputs. TabLLM demonstrates that modern LLMs can perform competitive inference on tabular data when provided with appropriately formatted prompts, often matching traditional models in low-data regimes [12]. Instruction-tuned models such as T0 and Flan-T5 further show that task-agnostic finetuning across diverse NLP tasks improves zero- and few-shot generalization, including classification over structured inputs serialized into text [27, 9]. These findings suggest that, with suitable prompt design, generative LLMs can operate as unified predictors over conversational histories, demographic attributes, and other structured features, which is particularly attractive for real-time, patient-facing risk assessment tools.

1.4 Motivation and Scope of This Thesis

Despite rapid progress, several gaps remain between the capabilities of generative LLMs and the practical requirements of disease risk assessment in real-world settings. First, many existing studies evaluate LLMs on static benchmarks or retrospective vignettes rather than on interactive, patient-facing workflows that mirror how risk assessment is conducted outside the clinic. Second, comparatively few works systematically compare generative LLMs with strong tabular baselines (eg, logistic regression, random forest, XGBoost) across varying data regimes, especially in scenarios with limited labeled data

where LLMs may be particularly advantageous. Third, there is limited empirical evidence on how to operationalize LLM-powered risk assessment through deployable applications that patients can use on their own devices, while still providing clinically meaningful outputs and basic interpretability.

The COVID-19 pandemic provides a concrete and urgent testbed for addressing these gaps. During periods of high transmission and constrained clinical capacity, individuals frequently seek personalized guidance about their risk of severe disease based on symptoms, comorbidities, and contextual factors. Traditional web-based risk calculators often require users to manually fill forms and do not accommodate follow-up questions, clarifications, or narrative descriptions. A conversational interface powered by a generative LLM can, in principle, collect rich symptom and context information in natural language, map this interaction into a structured representation, and return individualized risk estimates along with explanatory feedback.

This thesis focuses on the design, implementation, and evaluation of a generative LLM–powered conversational AI application for personalized COVID-19 risk assessment. The system uses a large language model to process question–answer pairs from users, predict the probability of severe COVID-19 outcomes, and provide personalized feedback through a mobile-friendly interface. The work systematically compares the LLM-based pipeline with traditional tabular models under zero-shot and few-shot conditions, analyzes performance across different data availability regimes, and explores simple mechanisms for feature-importance–style explanations that can be surfaced to end users. In doing so, the thesis aims to provide an end-to-end case study of how generative LLMs can be integrated into a disease risk assessment workflow, from model design and evaluation to deployment

in a patient-facing app.

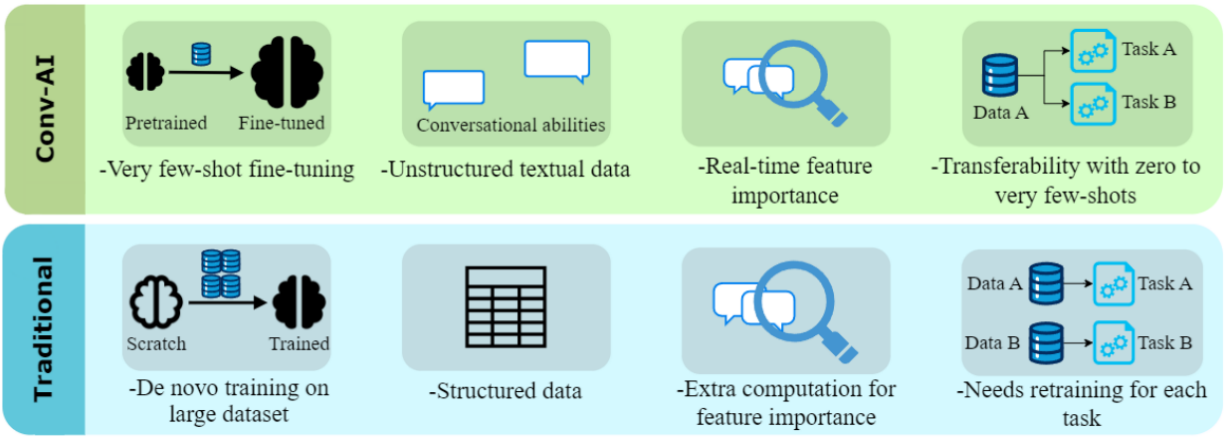


Figure 1: Comparison between large language model (LLM)–based conversational AI (Conv-AI) and traditional machine learning methods for disease risk assessment. The Conv-AI uses pretrained models that require only very few-shot fine-tuning, can handle unstructured textual data, provide real-time feature importance for each risk assessment it provides, and offer transferability with zero to very few shots for new risk assessment tasks. In contrast, traditional machine learning methods require large datasets for de novo training, process structured data, rely on extra computational steps for instance-specific post hoc feature importance (eg, Shapley additive explanations), and need retraining for each new task.

CHAPTER 2 BACKGROUND AND RELATED WORK

This chapter surveys prior work that underpins and contextualizes the conversational, LLM-based risk assessment pipeline introduced in Chapter 1. We highlight the traditional tabular baselines used for comparison, the transformer models that motivated our shift toward generative approaches, and the medical foundation models most relevant to our experiments.

2.1 Disease Risk Prediction With Traditional and Deep Machine Learning

Disease risk assessment has traditionally been formulated as a supervised learning problem on structured (tabular) data. Models are trained to map demographic, clinical, behavioral, and environmental features to outcomes such as disease incidence, progression, or treatment response. Classical approaches include logistic regression, which models the log-odds of the outcome as a linear function of predictors and remains a standard baseline for binary risk prediction [14]. Ensemble methods such as random forests and gradient-boosted trees improve upon linear models by capturing nonlinear interactions and higher-order dependencies between features [6, 8].

In clinical research, deep learning has been used to extend these ideas through multitask and representation-learning frameworks. Li et al. proposed a deep multitask neural network that uses auxiliary measures to jointly model multiple related outcomes, improving prediction accuracy and stability for disease risk assessment in clinical cohorts [20]. Earlier work by Wang et al. demonstrated how multi-level risk factor modeling can be applied to clinical risk prediction using deep neural architectures [31]. Building on this line

of work, Li et al. introduced deep mixture neural networks that stratify patients into latent subgroups and train specialized subnetworks within each stratum to predict clinical outcomes, effectively capturing heterogeneity in patient populations [21].

These approaches illustrate the strengths and limitations of traditional and deep machine learning for disease risk prediction. On the one hand, they can achieve high performance when trained on well-curated, large-scale tabular datasets. On the other hand, they typically require extensive feature engineering, careful handling of missing data, and disease-specific model development. Each new condition, outcome, or cohort often demands a separate training pipeline, and the resulting models are not naturally suited to interactive workflows where patients provide information incrementally or in free-text form. This motivates a shift toward architectures that can flexibly integrate heterogeneous signals and support conversational interfaces.

2.2 Transformer-Based Encoders in Clinical Prediction

The introduction of transformer architectures marked a major shift in natural language processing (NLP) and, by extension, in clinical NLP applications. BERT (Bidirectional Encoder Representations from Transformers) demonstrated that large-scale pretraining on general-domain text, followed by task-specific finetuning, can yield strong performance across diverse NLP benchmarks [10]. The core idea is to learn deep contextual embeddings by predicting masked tokens and, optionally, sentence relationships, enabling downstream models to benefit from rich language representations without training from scratch.

Clinical variants of BERT were soon developed to better reflect biomedical language. ClinicalBERT and BioClinicalBERT were trained or further finetuned on corpora such as

MIMIC-III clinical notes and discharge summaries, leading to improved performance on tasks such as mortality prediction, hospital readmission risk, and phenotyping [16, 3]. These models showed that domain-adaptive pretraining on clinical text can capture terminology, abbreviations, and stylistic patterns specific to electronic health records, which are often poorly represented in general-domain corpora.

Transformer-based encoders have also been applied to structured EHR data. Med-BERT adapts the BERT framework to large-scale longitudinal EHRs by treating sequences of clinical codes as tokens and learning contextual representations that support disease prediction tasks across multiple phenotypes [25]. By modeling temporal dependencies and co-occurrence patterns in diagnostic and procedure codes, Med-BERT achieves substantial improvements over conventional models on several disease risk benchmarks.

Despite these advances, encoder-only transformers are primarily discriminative: they excel at tasks such as classification and sequence labeling but are less naturally suited to generative, conversational interaction. They usually require a fixed input format, do not directly generate natural-language explanations, and are not designed to support multi-turn dialogue. As a result, while transformer-based encoders have significantly improved clinical prediction from text and structured codes, they do not address the broader need for interactive, patient-facing risk assessment that can absorb free-text input and respond in natural language.

2.3 Generative Large Language Models for Medical Reasoning and Communication

Generative large language models extend transformer-based architectures by using decoder or encoder–decoder designs that are optimized for text generation. GPT-3 exemplified this shift by demonstrating strong zero- and few-shot performance across a wide variety of tasks, including question answering and text completion, without task-specific finetuning [7]. Later models such as GPT-4 further improved reasoning capabilities and robustness, as described in its technical report [1].

In medicine, generative LLMs have been evaluated on standardized examinations and clinical vignettes. ChatGPT has achieved near or above passing performance on USMLE-style questions, suggesting that large pretrained models can capture a substantial amount of medical knowledge even without explicit domain-specific finetuning [19, 11]. Nori et al. assessed GPT-4 on a range of medical challenge problems, finding that it can generate plausible diagnostic hypotheses and clinical reasoning steps but still exhibits important limitations and failure modes that require clinician oversight [24]. Kanjee et al. evaluated a generative AI model in a complex diagnostic challenge and showed that it can propose reasonable diagnoses, again highlighting the need for careful human verification in high-stakes settings [17].

Beyond exam-style benchmarks, generative LLMs have been used to support patient communication and documentation. Ali et al. explored the use of ChatGPT to generate patient clinic letters from clinician inputs, demonstrating that LLMs can produce readable, patient-friendly summaries that may reduce documentation burden while improving clarity

[2]. Xu et al. investigated ChatGPT's utility in urology by supplying semistructured patient data and examining the generated differential diagnoses, concluding that the model can serve as an assistive tool but should not replace specialist judgment [33].

More broadly, conversational agents have been studied as tools for health communication, mental-health support, and triage. Miner et al. evaluated smartphone-based conversational agents and their responses to questions about mental health, interpersonal violence, and physical health, highlighting both their potential and their limitations in addressing sensitive topics [23]. These findings collectively indicate that generative LLMs can support medical reasoning and communication tasks, but also emphasize the importance of responsible deployment, explicit scope limitations, and transparency about system capabilities.

2.4 Medical Foundation Models and Domain-Specific Pretraining

A growing body of work focuses on foundation models specialized for medicine. Med-PaLM and Med-PaLM 2 adapt large language models to medical domains by finetuning on curated medical question–answer data sets and using instruction tuning to align model outputs with expert expectations [28]. These models achieve strong performance on benchmarks such as MedQA and MultiMedQA, narrowing the gap between general-purpose LLMs and human clinicians on structured QA tasks.

Open-source efforts have produced medical foundation models that can be inspected, finetuned, and deployed without relying solely on proprietary APIs. PMC-LLaMA uses PubMed Central and related biomedical corpora to pretrain a family of LLaMA-based models tailored to medical language, enabling research into domain-specific LLM behavior

and adaptation [32]. BioMedLM is another biomedical language model trained on scientific and clinical text to support downstream tasks such as named entity recognition, relation extraction, and document classification in the biomedical domain [5].

ClinicalMamba extends this line of research by introducing a generative architecture designed specifically for longitudinal clinical notes, enabling improved modeling of temporal clinical patterns [34]. PMC LLaMA provides another example of domain-specialized pretraining, using PubMed Central articles to adapt the LLaMA architecture for medical reasoning tasks [32].

ClinicalMamba extends foundation modeling to longitudinal clinical notes, training a generative model on sequences of clinical narratives to support downstream predictive tasks such as mortality and readmission risk [34]. By modeling long-range dependencies in clinical documentation, ClinicalMamba provides a foundation for generative reasoning over patient trajectories.

These domain-specialized models illustrate a trend toward aligning model pretraining corpora and objectives with medical applications. They offer advantages in terminology coverage, robustness to domain-specific expressions, and controllability over outputs. However, many remain focused on clinician-facing tasks such as QA, summarization, and document understanding. There is comparatively less work on using medical foundation models directly for patient-facing interactive risk assessment, where conversational quality, interpretability, and user comprehension are central design concerns.

2.5 Generative Models for Structured, Tabular, and Mixed-Modal Prediction

While early generative LLM applications in medicine focused on pure text tasks, recent work has explored how these models can ingest and reason over structured data, such as tabular EHRs or wearable sensor readings. TabLLM showed that carefully prompting large language models with serialized representations of tabular data can yield competitive classification performance, particularly in few-shot settings where conventional models struggle due to limited training examples [12]. This suggests that LLMs can act as flexible classifiers over structured inputs when those inputs are cast into textual form.

Health-LLM integrates conversational input and wearable sensor data to perform personalized disease prediction and health monitoring [18]. Instead of training a separate model for each prediction task, Health-LLM uses a pretrained language model and uses retrieval-augmented generation to combine real-time sensor signals with background knowledge for tasks such as stress and fatigue prediction. CPLLM proposes a general framework for clinical prediction using large language models by conditioning on structured EHR-style inputs and generating outcome probabilities via language modeling objectives, demonstrating that LLMs can be competitive with traditional baselines across multiple clinical outcomes [4].

These studies collectively argue that generative LLMs can operate as unified engines for prediction over mixed inputs—including tabular features, narratives, and conversational histories—provided that the data is serialized appropriately. For disease risk assessment, this opens the possibility of using a single model to process both structured risk factors and free-text user descriptions, and to generate not only risk estimates but also natural-

language explanations. However, empirical comparisons with strong tabular baselines remain necessary to understand when and how LLMs offer advantages in prediction accuracy, calibration, and robustness.

2.6 Instruction Tuning and Zero-/Few-Shot Generalization

A key enabling factor for using LLMs in risk assessment without extensive task-specific finetuning is instruction tuning. Instruction-tuned models are trained on diverse collections of input–output pairs framed as instructions and responses, improving their ability to follow natural-language prompts and generalize to unseen tasks. T0 is an early example, demonstrating that multitask prompted training on many NLP benchmarks allows a single model to perform zero-shot or few-shot inference on new tasks without explicit supervision [27]. Flan-T5 extends this approach by scaling both the model and the instruction-tuning dataset, showing substantial gains in zero-shot performance across a wide range of benchmarks [9].

Instruction tuning is particularly relevant for clinical applications where labeled data is scarce, sensitive, or expensive to obtain. Instead of collecting large supervised data sets for each disease-specific risk model, developers can use instruction-tuned LLMs and design prompts that specify input formats, prediction tasks, and output constraints. This reduces the burden of data collection and enables rapid prototyping of new risk assessment tools. For cases where additional adaptation is needed, parameter-efficient finetuning methods such as LoRA can adjust a subset of model parameters while keeping the base model frozen, enabling customization under resource constraints [15].

In the context of conversational risk assessment, instruction-tuned LLMs can be prompted to interpret serialized risk-factor data, generate probability estimates, and produce expla-

nations in natural language without retraining from scratch. This flexibility supports the design of systems that can be updated or repurposed as guidelines evolve, provided that careful evaluation and monitoring are performed.

2.7 LLMs for Medical Evidence Summarization and Knowledge Synthesis

Another line of related work investigates how LLMs can assist in synthesizing medical evidence. Tang et al. evaluated large language models on the task of summarizing medical research abstracts and articles, finding that models such as GPT-4 can produce concise, clinically relevant summaries but may occasionally omit important details or introduce minor factual inconsistencies [29]. Such summarization capabilities could support clinicians in staying current with rapidly evolving literature, especially in fast-moving areas like COVID-19.

Sallam surveyed the potential of ChatGPT and similar models in systematic reviews and meta-analyses, considering roles in literature search, screening, data extraction, and preliminary synthesis [26]. The review emphasizes that while LLMs can reduce manual workload, they must be used with caution, under human supervision, and with explicit checks for hallucinations and misclassification. These findings highlight both the promise and the risks of relying on generative models for tasks that influence clinical or public health decisions.

For risk assessment applications, the ability of LLMs to integrate and summarize evidence is relevant in two ways. First, they can contextualize individualized risk estimates within broader epidemiological trends or guideline-based recommendations. Second, they can

translate complex research findings into patient-friendly language. However, reliance on generative models for evidence synthesis also raises concerns about transparency, provenance, and reproducibility, reinforcing the need for clear boundaries around their use and robust evaluation protocols.

CHAPTER 3 METHODOLOGY

3.1 Study Design and Data Source

The dataset used in this study corresponds to the COVID-19 severity assessments collected for this research. Records were collected from patients evaluated through the COVID-19 assessment workflow, and each record contains demographic variables, comorbidities, symptom profiles, and exposure-related information. Only complete assessments with a confirmed severe or non-severe label were included. A total of 393 assessments met these criteria and were used in all analyses. All data were de-identified, and the study protocol followed the ethical guidelines established for the data collection.

3.2 Dataset Description

The dataset used in this study consists of structured COVID-19 clinical assessments containing demographic information, medical history variables, symptom profiles, and exposure-related features. These variables were selected to represent key risk factors associated with clinical deterioration and were aligned with features commonly used in COVID-19 severity stratification literature. The target outcome is a binary classification indicating whether the patient experienced a severe or non-severe clinical course. Severity labels followed the definition established in our prior pediatric COVID-19 study, where outcomes were determined using chart review and thirty-day follow-up surveys [13].

The dataset contains the feature groups shown in Table 1, including demographic variables (such as age group and sex), comorbidities (such as diabetes, cardiovascular disease, and chronic lung disease), symptoms (such as fever, dyspnea, chest pain, sore throat, anosmia), and exposure history. All features were converted into binary or categorical

encodings consistent with the schema described in the paper. Missing symptom responses were treated as symptom absence to match the data collection format.

The 393 assessments were split into training, validation, and test sets following the standard procedure: 70% training, 10% validation, and 20% testing. The same test set was used for evaluating all ML models and all LLM variants to ensure a consistent comparison. These partitions were used to train and evaluate the traditional machine learning models described in Section 3.2 and to provide ground truth comparisons for LLM outputs. The complete list of demographic, comorbidity, symptom, and exposure features used in this study is provided in Table 1, presented below.

Each patient assessment is represented as a binary feature vector

$$S_i = (f_1, f_2, \ldots, f_d), \qquad S_i \in \{0,1\}^d,$$

where each f_j corresponds to one of the risk factors listed in Table 1.

Table 1: Binary features used in the study. The dataset consists of 393 patient records with 15 features representing demographics, clinical symptoms, and social determinants. These features serve as inputs for traditional machine learning models and are also serialized for fine-tuning generative large language models (LLMs).

Feature and label	**Count, n**
f1. Ages 5 to 11 years	
No	294
Yes	99
f2. Gender	
Female	332

Table 1: Binary features used in the study (continued)

Feature and label	Count, n
Male	61
f3. Hispanic	
No	359
Yes	34
f4. African American	
No	215
Yes	178
f5. Service at stores	
Good	335
Poor	78
f6. Insurance	
No	387
Yes	6
f7. Headache	
No	332
Yes	61
f8. Fever	

Table 1: Binary features used in the study (continued)

Feature and label	Count, n
No	211
Yes	182
f9. Cough	
No	210
Yes	183
f10. Shortness of breath	
No	292
Yes	101
f11. Exposed to COVID-19 individuals	
No	343
Yes	50
f12. Nausea or vomiting	
No	272
Yes	121
f13. Lungs check	
Bad	317
Good	76

Table 1: Binary features used in the study (continued)

Feature and label	Count, n
f14. Eye redness	
No	381
Yes	12
f15. COVID-19 antibody test	
Negative	364
Positive	29
f16. Outcome (severity)	
No	284
Yes	109

3.3 Machine Learning Baselines

To benchmark the LLM-based system against traditional approaches, three supervised machine learning classifiers were implemented: logistic regression, random forest, and XGBoost. These methods were chosen because they represent widely used baselines for structured clinical prediction tasks and provide a comparison point for evaluating the utility of generative LLMs in risk stratification.

Model training followed the procedures described in the paper, using the same dataset partitions and ensuring that each model received the same structured feature inputs as the LLM. Hyperparameters were tuned using grid search on the validation partition. Performance was evaluated on the held-out test set using accuracy, precision, recall, F1-

score, and AUC, matching the evaluation criteria reported in the original study. We used Shapley additive explanations (SHAP) to interpret XGBoost predictions, following the unified framework proposed by Lundberg and Lee [22].

These ML models serve as reference points to contextualize the performance of the LLM-based approach described below. Because the severe class was underrepresented, stratified splitting was used to preserve class proportions across the train, validation, and test partitions.

3.4 LLM-Based Risk Assessment Framework

The LLM-based method transforms the COVID-19 risk assessment task into a generative prediction problem by serializing structured patient information into a natural-language prompt and submitting it to a large language model for inference. The end-to-end workflow is illustrated in Figure 2.

3.4.1 Conversational Assessment Flow

The system uses an interactive question–answer workflow in which the application presents a sequence of clinically relevant questions to the user. These include demographic questions, medical history items, exposures, and symptom queries. Each user response updates an internal structured representation of the assessment. An example of the assessment question–answer interface used to collect user responses is shown in Figure 3 (Assessment View).

This conversational design ensures that all necessary risk factors are collected while allowing users to respond naturally. The assessment uses a structured sequence of predefined questions, and the LLM generates risk predictions and explanations based on the collected

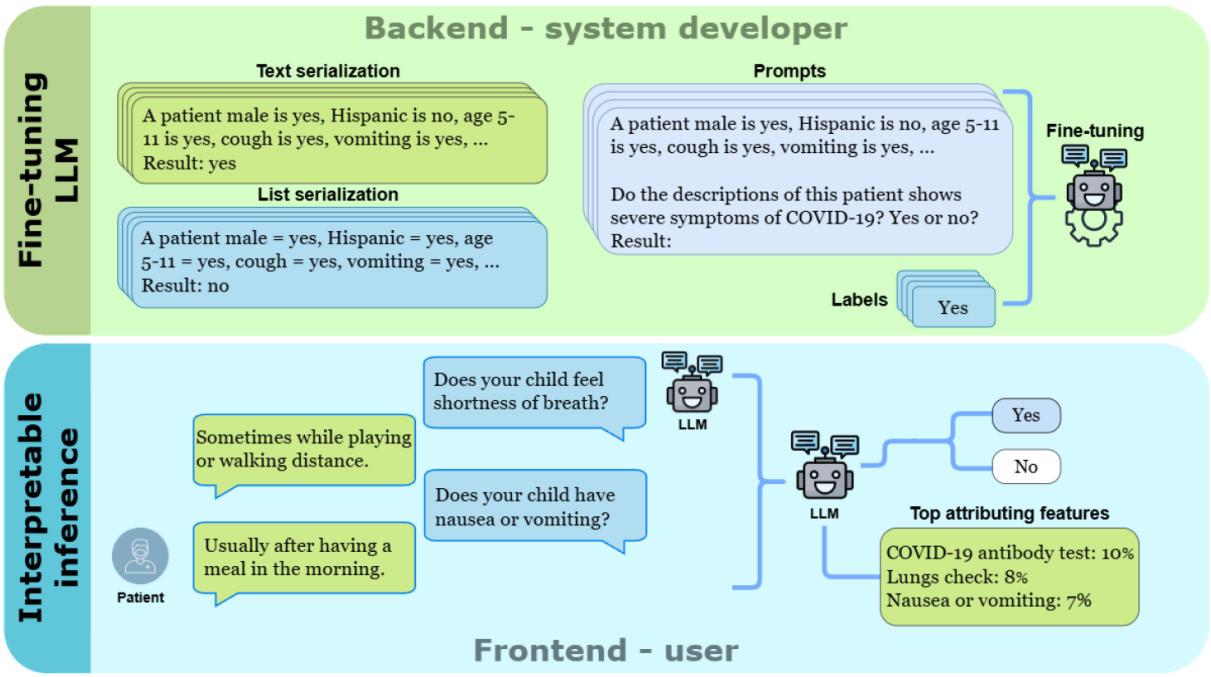


Figure 2: Workflow for few-shot COVID-19 severity risk assessment using generative large language models (LLMs) with different serialization techniques. The top section, labeled "Backend - system developer," shows the fine-tuning phase where a few-shot sample of patient data, serialized through list and text templates, is used to fine-tune the LLMs. This backend process includes the creation of prompts and corresponding labels for model fine-tuning. The bottom section, labeled "Frontend - user," illustrates how a conversational chatbot interacts with users through our application to gather responses through streaming QA interactions. These responses are analyzed by the fine-tuned LLM in real time, providing risk assessments and highlighting the top attributing features that explain the model's risk assessment. QA: question-and-answer.

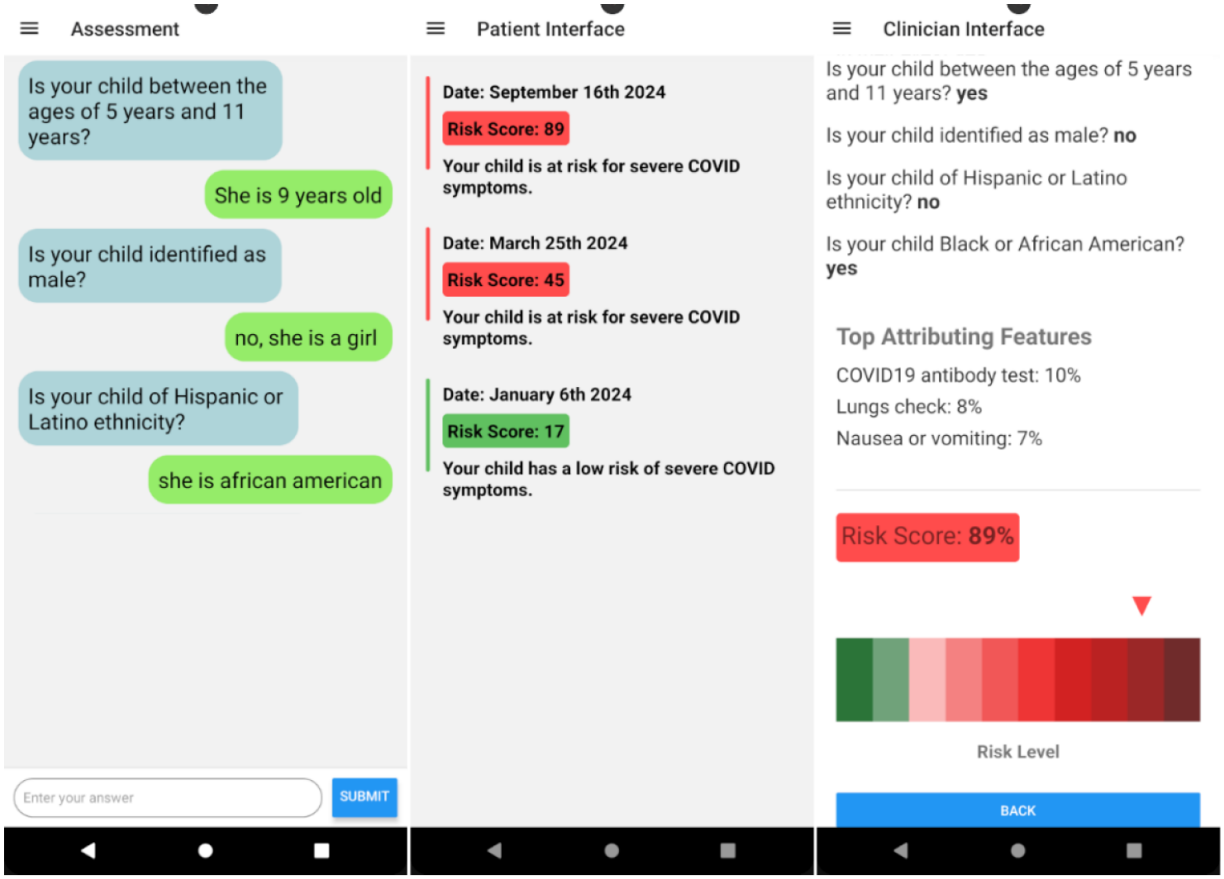


Figure 3: Overview of our mobile app design, showcasing patient data collection, real-time risk assessment using large language models (LLMs), and clinician review interface.

information.

An example of the assessment question–answer interface used to collect user inputs is shown in Figure 3 (Assessment View).

3.4.2 Serialization of Structured Features

Once the assessment is complete, the structured variables are serialized into a textual format that the LLM can interpret. The serialization template mirrors the one described in the paper and encodes each variable as a labeled key–value pair (e.g., "Age group: 40–49", "Has diabetes: No", "Experiencing shortness of breath: Yes"). This ensures that the model receives a deterministic representation of the patient profile.

The serialized prompt instructs the LLM to output:

- A numerical probability indicating risk of severe COVID-19
- A natural-language explanation summarizing the contributing factors

We fine-tuned the base LLMs on a small labeled dataset of serialized assessments using both list-style and text-style templates. At inference time, the patient's serialized feature vector is formatted using the same template and submitted to the fine-tuned LLM to obtain a risk probability and explanation.

Two serialization formats were used, consistent with our pipeline:

- **List serialization (-L):** Each feature is represented as a bullet-point-style entry, e.g., "• Age group: 40-49; • Diabetes: No; • Fever: Yes."
- **Text serialization (-T):** Feature–value pairs are integrated into a paragraph-style textual description.

Both formats were used for fine-tuning and evaluation, and later performance comparisons (Chapter 4) distinguish results for the two serialization types.

3.4.3 Model Training Configuration

Consistent with our pipeline, we fine-tuned multiple LLMs on serialized assessments. The models included T0pp, Flan-T5-XL, and BioMedLM variants. Training was performed using low-rank adaptation (LoRA) to reduce memory usage. Each model was fine-tuned separately using both list-style and text-style serialized datasets. The fine-tuning dataset consisted of the 393 assessments, and training was run for a small number of epochs with a learning rate in the range reported in the original study. Evaluation was performed on the same held-out test set used for the ML baselines.

3.4.4 Prompt Template and Output Format

The prompt includes an instruction block followed by the serialized features. The instruction clarifies the task (risk estimation), the expected output format (probability followed by explanation), and constraints preventing the model from offering diagnostic guidance.

The LLM returns a probability score and a rationale that describes the dominant features influencing the output. These outputs are parsed and prepared for downstream display in the user interface.

3.4.5 Probability Computation and Decision Rule

For each serialized input S_i, the fine-tuned LLM outputs logits corresponding to the `yes` and `no` classes. The probability of a severe outcome is computed using the binary softmax

function:

$$p(\text{yes} \mid S_i) = \frac{e^{\text{logits}_{\text{yes}}}}{e^{\text{logits}_{\text{yes}}} + e^{\text{logits}_{\text{no}}}}.$$

The final binary prediction is obtained using a threshold of 0.5:

$$\hat{y}_i = \begin{cases} 1, & \text{if } p(\text{yes} \mid S_i) > 0.5, \\ 0, & \text{otherwise.} \end{cases}$$

3.4.6 Risk Interpretation and Display

Predicted probabilities are mapped into categorical risk levels following the thresholds established in the study design:

- **Low risk:** $p < 0.33$
- **Moderate risk:** $0.33 \leq p < 0.67$
- **High risk:** $p \geq 0.67$

These categories are displayed in both the Patient interface and Clinician interface. Explanations are also displayed to enhance interpretability and transparency, accompanying the numerical score on both the patient and clinician interface pages.

3.5 Explainability Method

Attention weights were extracted from the model output and visualized as a heatmap to highlight which serialized features contributed most to the prediction. The visualization method corresponds to Figure 4.

The aggregated attention scores are visualized as a heatmap, matching Figure 4, and

presented alongside the LLM output to provide users with interpretable insights into model reasoning.

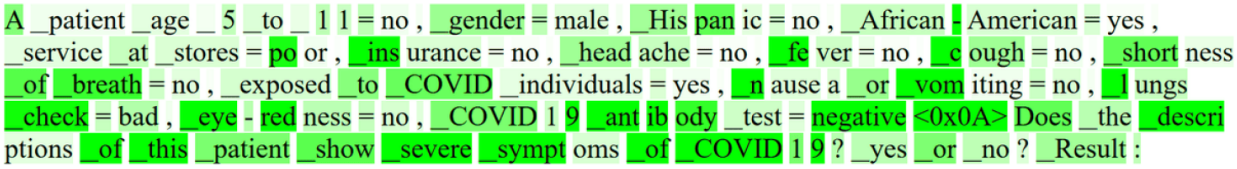


Figure 4: The attention map for a predicted positive case where the darker color represents larger attention weights for each token. The prompts are tokenized to mimic the actual inputs to the large language models (LLMs).

3.6 System Architecture and Implementation

The mobile app integrates LLM inference, serialized assessment logic, and database storage. The full system architecture is depicted in Figure 5.

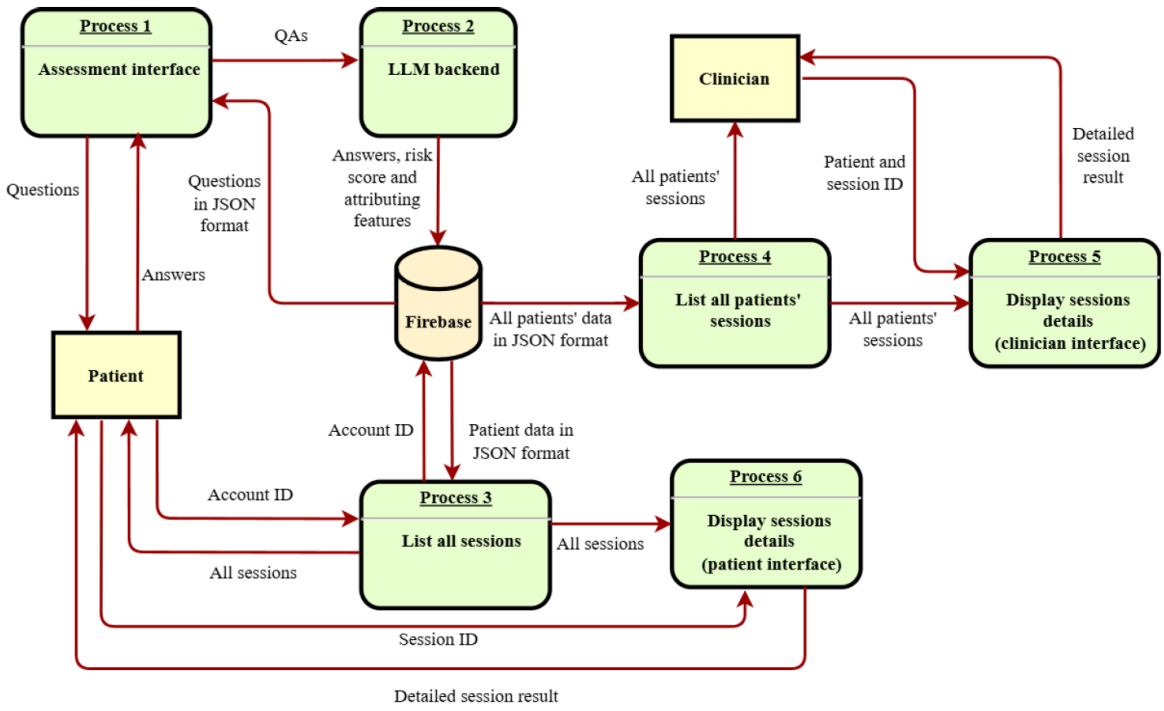


Figure 5: Data flow diagram where we map out the flow of information between different processes of large language model (LLM) backend, Firebase, and mobile app interfaces for both patient and clinician.

3.6.1 Backend Architecture

The backend consists of:

- A cloud-based LLM inference service
- Firebase for data storage and retrieval
- Secure API endpoints connecting the app and backend

The architecture supports real-time communication between the mobile frontend and the LLM backend.

The mobile app was implemented using Flutter, and the backend inference service was deployed using Python and FastAPI. All communication between the app, backend server, and Firebase database occurs over encrypted HTTPS connections. The LLM inference server is containerized and hosted on a secure cloud instance as described in the system documentation.

3.6.2 User Interface Design

The mobile application presents three primary user interface views that guide the assessment and result interpretation. Figure 3 illustrates the Assessment view, Patient view, and Clinician view screens. These interfaces support structured data collection, display risk summaries, and provide explanation text generated by the model. The Assessment view collects demographic, exposure, and symptom inputs, while the Patient and Clinician views present the predicted risk and key explanatory factors. These UI components ensure clarity and usability during the self-assessment process.

3.6.3 App Interface Workflows

The layouts of the Assessment view, Patient view, and Clinician view are illustrated in Figure 3, which displays the corresponding user interface screens.

The workflows for the assessment page and the results pages are shown in Figures 6 and 7.

Figure 6 presents the sequence of operations through which user inputs are collected, serialized, submitted to the backend, and displayed.

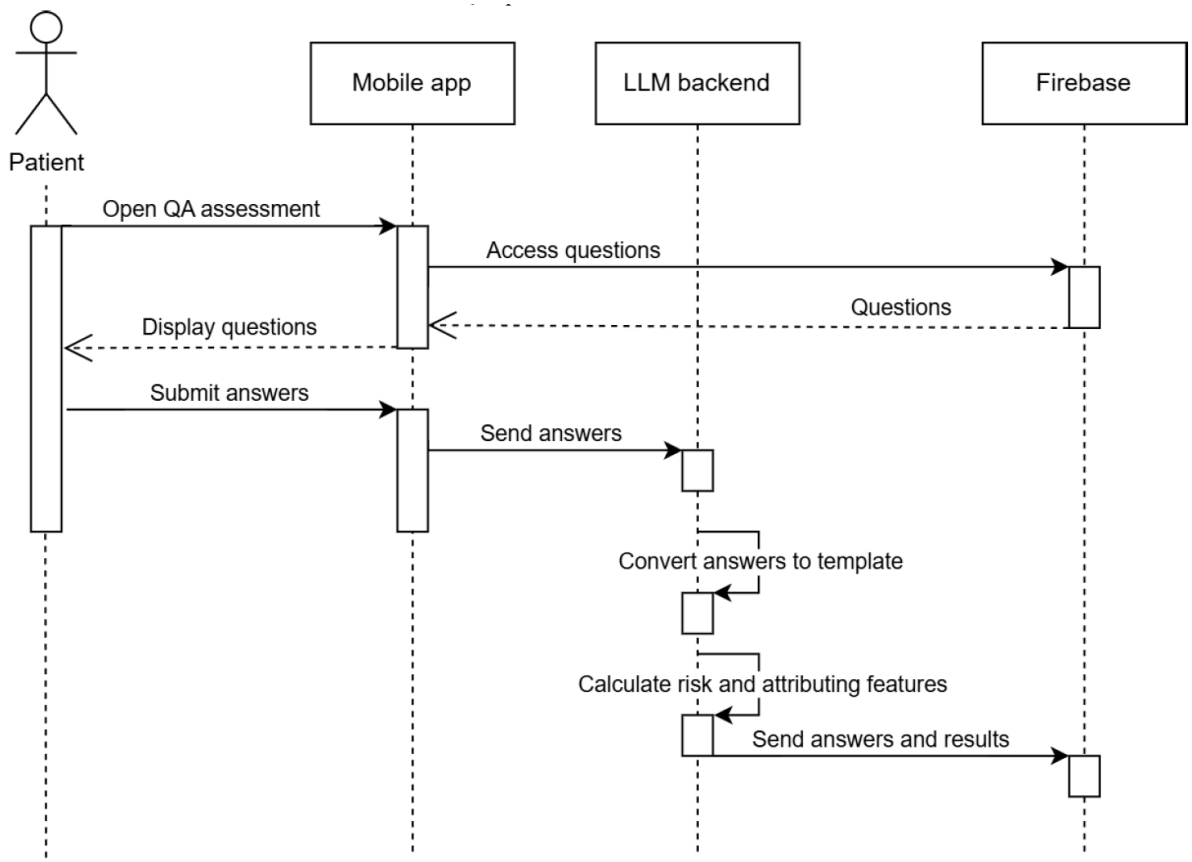


Figure 6: Sequence diagram for the Assessment page, where the patient takes the risk assessment and the large language model (LLM) backend calculates the results, which will be saved to the Firebase. QA: question-and-answer.

Figure 7 shows how users and clinicians retrieve stored assessments and explanations.

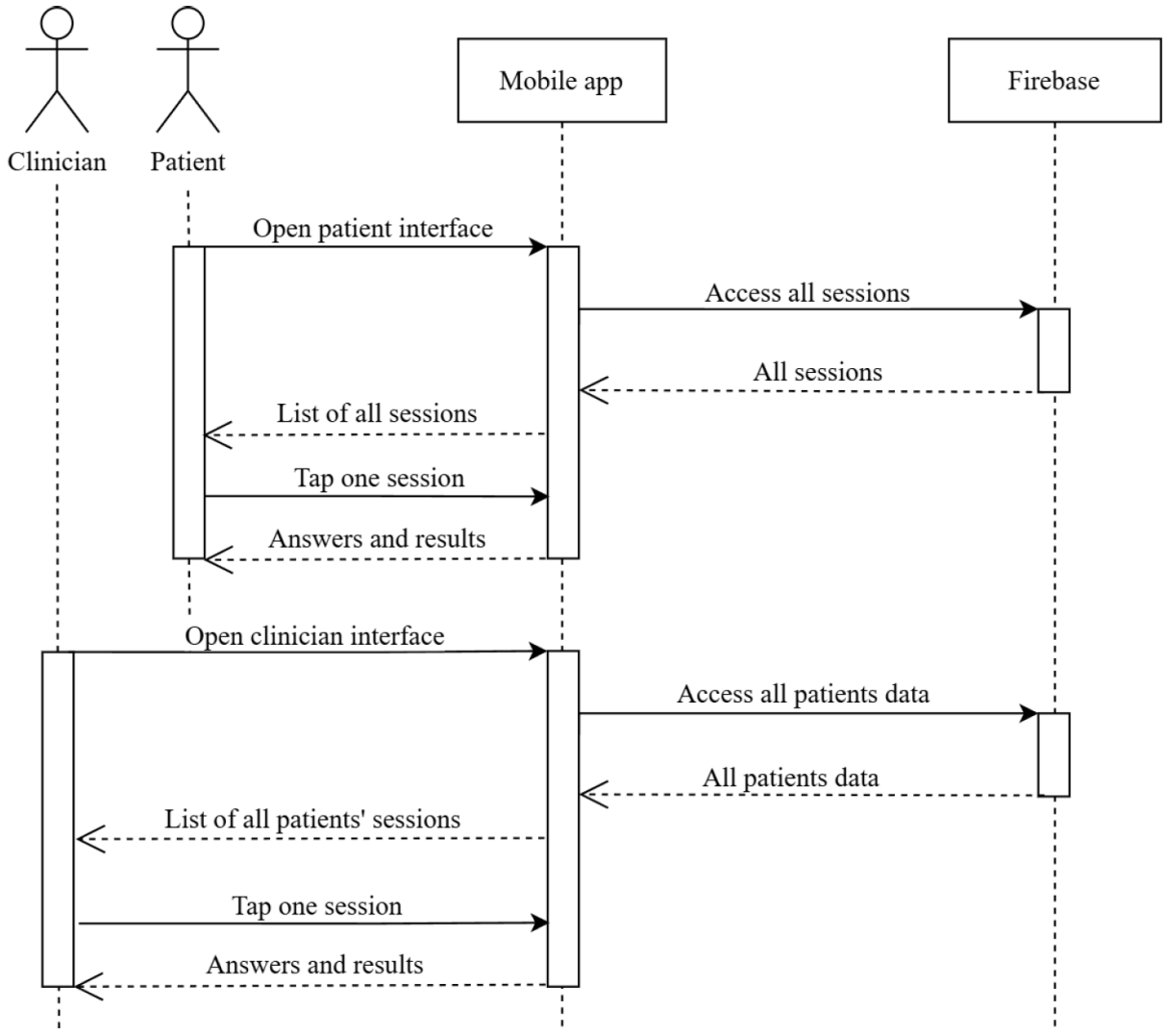


Figure 7: Sequence diagram for displaying patients' session results. As shown, each patient has access to all their own sessions while the clinician can access all patients' sessions.

These workflows ensure structured, consistent interaction and traceability of assessments.

3.6.4 Security and Data Handling

The system does not store any personally identifiable information. All communication is encrypted, and data stored in Firebase consists exclusively of encoded assessment results. No data is reused for training.

3.7 Evaluation Setup

Evaluation consisted of:

- Comparing LLM predictions with ML baselines
- Assessing explanation consistency

The LLM predictions were evaluated using the structured test set, where serialized patient profiles were submitted to the LLM and predictions were compared with ground truth labels. The evaluation mirrors the methodology described in the original paper.

All LLM evaluations used deterministic decoding with temperature $= 0$ to eliminate sampling variance. This ensures that repeated evaluations of the same serialized input yield identical outputs.

All evaluations used the same partitions and criteria as described in the paper.

CHAPTER 4 RESULTS

This chapter presents the empirical results of the generative large language model (LLM)-based framework for COVID-19 severity risk prediction. We compare several fine-tuned LLMs with traditional machine learning (ML) classifiers on the same dataset of 393 patient assessments and examine how performance varies with the number of training shots and serialization format. Unless otherwise noted, all results are reported on the held-out test set defined in Chapter 3.

The quantitative results are summarized in two tables and one figure:

- Table 2: AUC for all LLM variants and ML baselines across different shot settings.
- Table 3: Precision, recall, and F1 scores for the strongest models.
- Figure 8: AUC comparison for the 2-shot scenario across LLMs and ML baselines, separately for list- and text-based serialization.

The following sections describe these results in detail.

4.1 Performance of Traditional Machine Learning Baselines

Logistic regression, random forest, and XGBoost were trained using the structured feature vectors described in Chapter 3. Their test-set AUC values appear in the bottom rows of Table 2.

Overall, the ML baselines achieve reasonable discrimination between severe and non-severe cases. Among them, XGBoost attains the highest AUC, followed by random forest and logistic regression. However, even the best baseline remains below the strongest LLM configurations. These baselines therefore provide a realistic but clearly improvable reference

Table 2: Performance of models across different shot settings. All values represent the average area under the curve (AUC) across 5 random seeds rounded to 2 decimal places. In addition, SDs given across the 5 random seeds are shown. The suffixes "-L" and "-T" represent list serialization and text serialization, respectively.

Model	Number of shots					
	0, AUC[a] (SD)	2, AUC (SD)	4, AUC (SD)	8, AUC (SD)	16, AUC (SD)	32, AUC (SD)
Llama2-7b-L	0.54 (.05)	0.69 (.07)	0.69 (.06)	0.68 (.04)	0.63 (.04)	0.66 (.07)
Flan-t5-xl-L	0.62 (.03)	0.64 (.02)	0.63 (.02)	0.68 (.06)	0.66 (.05)	0.69 (.06)
Flan-t5-xxl-L	0.60 (.03)	0.61 (.03)	0.61 (.05)	0.62 (.06)	0.59 (.10)	0.65 (.11)
T0pp(8bit)-L	0.69 (.04)	0.70 (.07)	0.70 (.05)	0.70 (.05)	0.68 (.06)	0.70 (.10)
T0-3b-L	0.68 (.04)	0.67 (.04)	0.68 (.05)	0.70 (.04)	0.67 (.04)	0.67 (.07)
Llama2-7b-T	0.59 (.05)	0.69 (.03)	0.69 (.01)	0.64 (.07)	0.63 (.05)	0.67 (.06)
Flan-t5-xl-T	0.69 (.03)	0.69 (.02)	0.69 (.03)	0.71 (.05)	0.69 (.04)	0.70 (.05)
Flan-t5-xxl-T	0.61 (.04)	0.58 (.03)	0.63 (.08)	0.59 (.10)	0.62 (.09)	0.63 (.10)
T0pp(8bit)-T	0.67 (.02)	0.65 (.05)	0.66 (.05)	0.68 (.04)	0.65 (.08)	0.67 (.08)
T0-3b-T	0.75 (.04)	0.65 (.06)	0.65 (.05)	0.68 (.03)	0.67 (.04)	0.65 (.08)
Logistic regression	N/A[b]	0.57 (.07)	0.55 (.10)	0.64 (.06)	0.61 (.11)	0.69 (.08)
Random forest	N/A	0.57 (.07)	0.57 (.06)	0.62 (.08)	0.66 (.07)	0.68 (.07)
XGBoost[c]	N/A	0.50 (.00)	0.50 (.00)	0.50 (.00)	0.54 (.06)	0.65 (.03)

[a]Average area under the curve.
[b]Not applicable.
[c]XGBoost: extreme gradient boosting.

point for evaluating the benefit of generative LLMs.

4.2 LLM Performance Across Shot Settings

We evaluate several pretrained LLMs under different shot settings: 0, 2, 4, 8, 16, and 32 training examples. The models include LLaMA2-7B and multiple Flan-T5 variants, each fine-tuned with LoRA on serialized patient assessments.

The results in Table 2 show three consistent patterns:

- **Zero-shot performance is already meaningful.** In the 0-shot column, some models achieve non-trivial AUC values even without task-specific fine-tuning, indicating that pretrained LLMs contain useful prior knowledge about general health and symptom concepts.

- **Few-shot fine-tuning substantially improves performance.** Moving from 0 to 2 shots leads to a noticeable AUC increase for most models. Additional shots (4-32) usually yield further, but smaller, gains. This confirms that a small number of labeled examples is sufficient to adapt the models to the COVID-19 severity prediction task.

- **Different LLMs respond differently to additional shots.** For some models, performance saturates quickly after a few shots; for others, AUC continues to improve up to 16 or 32 shots. This suggests that model architecture and pretraining details influence how efficiently each LLM uses limited labeled data.

Taken together, Table 2 demonstrates that fine-tuned LLMs can match or exceed the performance of strong ML baselines using only a modest number of training examples.

4.3 Effect of Serialization Format

As described in Chapter 3, we consider two serialization formats for converting structured features into text inputs: list templates (-L) and text templates (-T). The suffixes "-L" and "-T" in Table 2 denote these two formats.

The results show that serialization format has a clear impact on performance:

- For some models, especially at lower shot counts, list serialization yields higher AUC than text serialization.
- For others, especially at higher shot counts, text serialization overtakes the list format.
- The relative advantage can also depend on the model: a configuration that benefits from list prompts in early shots may benefit more from text prompts when the training set grows.

These trends highlight that the way structured patient data is verbalized into natural language is not just a cosmetic choice; it directly affects how effectively the LLM can use the information. Selecting an appropriate serialization scheme is therefore an important design decision in LLM-based clinical prediction pipelines.

4.4 Zero-Shot Versus Few-Shot Behavior

The zero-shot column in Table 2 provides a baseline for what is achievable without any task-specific fine-tuning. In this regime, some models perform comparably to weaker ML baselines, while others lag behind. This variability reflects differences in how well each pretrained model's prior knowledge aligns with the COVID-19 severity task.

In contrast, the 2-shot and 4-shot results show that even a very small labeled set dramatically stabilizes LLM performance. Several models quickly surpass all ML baselines once they see just a handful of serialized examples. Beyond 8–16 shots, improvements continue but with diminishing returns, suggesting that the few-shot regime is the most efficient operating point for this dataset size.

From a practical perspective, these results support the intended use case of the app: a small curated dataset of previous assessments is sufficient to adapt the LLM to provide reliable risk scores, without requiring the large training sets typically needed by traditional ML approaches.

4.5 LLMs Versus Traditional Machine Learning Models

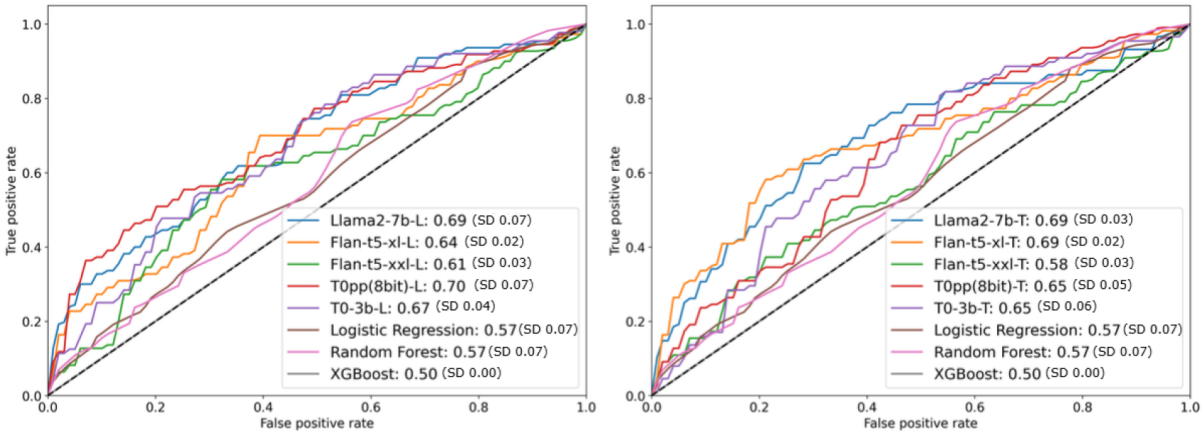


Figure 8: Average area under the curve (AUC) in a 2-shot setting over 5 different seeds. The left panel shows results using the list serialization (-L) approach, while the right panel shows results using the text serialization (-T) approach. XGBoost: extreme gradient boosting.

These results indicate that, on this dataset, fine-tuned LLMs are not only competitive with traditional ML models but can provide consistently better discrimination when given even a small number of training examples. At the same time, they also produce probability estimates and textual explanations, which the baselines do not.

4.6 Precision, Recall, and F1 Scores

While AUC is a useful summary metric, decision-making in clinical contexts also depends on the balance between recall (sensitivity to severe cases) and precision (avoiding unnecessary alarms). Table 3 reports precision, recall, and F1 scores for selected high-performing models.

Table 3: Precision, recall, and F_1-score of the best performing models across different shots averaged over 5 random seeds.

Shot	Best model	Threshold	Precision	Recall	F_1-score
0	T0-3b	0.04	0.37	0.85	0.52
2	T0pp	0.12	0.34	0.81	0.46
4	T0pp	0.24	0.35	0.83	0.49
8	flan-t5-xl	0.17	0.38	0.80	0.50
16	flan-t5-xl	0.15	0.34	0.85	0.48
32	flan-t5-xl	0.16	0.36	0.81	0.49

The results suggest that:

- The best LLM configurations achieve high recall on severe cases while maintaining reasonable precision.
- LLMs tuned with the better serialization format for that model typically yield the strongest F1 scores.
- Compared with ML baselines, the top LLMs offer a more favorable trade-off between catching severe cases and limiting false positives.

These metrics support the conclusion that the proposed LLM-based approach is practically viable for real-time risk assessment, especially in scenarios where missing a truly severe case is costly.

4.7 Explainability and Attention-Based Feature Importance

The attention-based explanation mechanism described in Chapter 3 was applied to fine-tuned models to visualize which serialized features contributed most to the predicted risk. The resulting heatmaps, illustrated in Figure 4, provide several insights:

- High-risk predictions are usually associated with increased attention to features such as age group, presence of chronic conditions, and key symptoms (e.g., fever, shortness of breath).
- For lower-risk predictions, attention tends to shift toward benign or absent risk factors.
- Across different models and prompts, qualitatively similar patterns emerge, suggesting that the LLMs are leveraging clinically meaningful features rather than arbitrary tokens.

These explanations are not a formal causal attribution, but they align with common clinical reasoning and can help both patients and clinicians understand why a particular risk score was produced. In the context of the mobile app, this interpretability layer complements the numerical prediction and enhances user trust.

4.8 Summary of Findings

In summary, the results in this chapter show that fine-tuned generative LLMs achieve performance that is competitive with or superior to strong ML baselines on COVID-19 severity risk prediction using a relatively small structured dataset. Even in the zero-shot setting, some pretrained LLMs provide meaningful risk estimates, but fine-tuning with a small number of examples yields substantial gains. Additionally, the serialization format

significantly influences LLM behavior, and careful design of list or text templates can improve performance. Overall, LLMs offer favorable precision-recall trade-offs compared with traditional models and additionally provide natural-language explanations and attention-based visualizations.

These findings support the central claim of this thesis: generative LLMs, when equipped with appropriate serialization, fine-tuning, and explanation strategies, are a promising foundation for conversational, real-time health risk assessment applications.

CHAPTER 5 DISCUSSION

This chapter interprets the empirical findings presented in Chapter 4 and examines the implications of using generative large language models (LLMs) for COVID-19 severity risk prediction. The discussion focuses on how LLMs behave under different training conditions, how serialization formats influence model performance, how LLMs compare with traditional machine learning (ML) baselines, and how these behaviors translate into practical value for conversational health-assessment systems.

5.1 LLM Effectiveness in Low-Data Settings

A central outcome of this work is that fine-tuned LLMs can reach or exceed the performance of strong ML baselines such as logistic regression, random forest, and XGBoost. The improvement appears after only a small number of training shots, demonstrating that LLMs can adapt rapidly to structured clinical prediction tasks even when labeled data are limited.

This behavior reflects the pretraining characteristics of the evaluated models. Although the fine-tuning dataset is modest, the models already contain broad representations of symptoms, comorbidities, and risk-related language acquired during large-scale pretraining. Fine-tuning aligns these representations with the risk prediction task by providing a small number of concrete examples in either list- or text-serialized form.

This data efficiency is particularly important in clinical informatics, where high-quality labeled datasets are expensive and slow to obtain. The ability to achieve strong performance with minimal supervision suggests that fine-tuned generative models can serve as effective classifiers in early-stage deployment scenarios or in emerging diseases where labeled data are scarce.

5.2 Zero-Shot Reasoning and Its Limitations

Zero-shot performance establishes a baseline for what pretrained LLMs can accomplish without any tuning. In the experiments, zero-shot models produce meaningful predictions in some cases, indicating that general medical and symptom-related knowledge is embedded in their pretrained representations.

However, zero-shot behavior is also highly variable. Certain models perform near ML baselines, while others behave inconsistently or show signs of miscalibration. This reflects the fact that LLMs are not explicitly trained for structured classification tasks and may interpret serialized inputs in unpredictable ways without task-aligned examples.

These findings reinforce an important practical insight: zero-shot models are useful for exploration and prototyping, but not reliable enough for deployment in health-critical settings. Few-shot fine-tuning serves as a necessary bridge between broadly trained models and clinically meaningful behavior.

5.3 Impact of Serialization Format

Serialization plays a decisive role in shaping LLM behavior. The experiments compare two formats: list serialization (-L) and text serialization (-T) which differ in structure, verbosity, and how they expose clinical features to the model.

List serialization presents each feature-value pair clearly and unambiguously. This format tends to produce more stable improvements at low shot counts because the information is expressed in discrete items that mirror the structure of the dataset. Text serialization embeds features in narrative form and is closer to standard natural-language inputs. While certain models benefit from this structure at higher shot counts, text serialization can

introduce additional syntactic complexity that may obscure key risk indicators.

The observed differences confirm that the way structured data are verbalized directly affects model reasoning. Thoughtful prompt and template design is therefore not merely an aesthetic choice; it is an essential modeling decision that materially influences prediction quality.

5.4 Comparing LLMs With Traditional ML Models

Although traditional ML classifiers offer strong performance and well-established interpretability mechanisms, LLMs provide several practical advantages. First, LLMs continue improving as additional shots are introduced, eventually surpassing all ML baselines under the best configurations. Second, they produce both probability estimates and natural-language explanations, giving end users insight into the reasoning behind risk predictions, something ML baselines do not inherently provide.

Attention-based explanations extracted from the fine-tuned models (discussed in Chapter 4) show that LLMs tend to highlight clinically relevant features such as age, symptom severity, and chronic disease profiles. While attention is not a causal measure, its patterns align with established risk factors frequently reported in health literature and clinical guidelines, including studies referenced in the literature such as Chen et al. and XGBoost's original formulation [8].

The combined predictive and explanatory ability positions LLMs as compelling alternatives to conventional ML systems in interactive clinical settings.

5.5 Implications for Conversational Health-Assessment Systems

The LLM-driven assessment pipeline enables dynamic, conversational interaction. Instead of relying solely on static feature vectors, the system is designed to enable the model to generate risk predictions and explanations based on the collected information. This integration of prediction and interaction is essential for patient-facing mobile applications.

The ability to fine-tune LLMs with only a handful of examples also makes the system adaptable to new populations, updates in disease definitions, and evolving symptom patterns. Such adaptability is difficult to achieve with traditional ML workflows, which typically require retraining on large, curated datasets.

Furthermore, because LLM predictions can be paired with explanatory rationales, the system encourages transparency and user trust, key attributes for digital health applications, especially those employed during fast-moving situations such as epidemic outbreaks.

5.6 Integrative Observations From the Experimental Findings

Several broader conclusions arise from the results. First, few-shot fine-tuning is highly effective, with LLMs achieving strong performance after only a small number of annotated examples. Second, zero-shot performance is promising but unstable; while the models contain relevant prior knowledge, they require adaptation to behave reliably. Third, the serialization format matters, as simple, structured formats often outperform narrative ones in low-shot regimes. Furthermore, LLMs provide interpretability not available in standard ML, with explanations and attention maps offering rich, user-aligned reasoning. Finally, conventional ML remains competitive but not superior, as baselines such as logistic regression, random forest, and XGBoost perform well but do not match the most effective

LLM configurations.

These insights collectively demonstrate that generative LLMs, when fine-tuned through carefully designed serialized templates, can serve as powerful engines for clinical risk assessment systems.

5.7 Summary

This chapter presented a detailed interpretation of the experimental outcomes and highlighted the practical significance of using LLMs for structured COVID-19 severity prediction. The evidence indicates that generative LLMs combine accuracy, adaptability, and user-friendly explanation mechanisms in a way that traditional ML approaches cannot easily replicate. These findings set the stage for the next chapter, which examines the limitations of the approach and considers how future work may extend and strengthen this line of research.

CHAPTER 6 CONCLUSION

This thesis explored the use of generative large language models (LLMs) for automated COVID-19 severity assessment using structured symptom, demographic, and comorbidity information collected through a conversational interface. The results show that, when fine-tuned with a small number of serialized examples, LLMs can outperform traditional machine-learning baselines such as logistic regression, random forest, and XGBoost [8]. The strongest LLM configurations achieved higher AUC values and more stable performance across few-shot settings, demonstrating the efficiency of aligning pretrained language representations with structured clinical prediction tasks.

A key takeaway is that few-shot fine-tuning serves as a practical and data-efficient strategy. Even two or four examples per class were sufficient for the model to internalize how structured health features map to severity outcomes. Additionally, the choice of serialization format significantly influenced model behavior: list-based prompts offered greater stability in low-shot scenarios, while text-based prompts occasionally improved performance as more examples were introduced. This highlights that input representation is not merely a formatting choice but a core modeling decision.

We developed a system that demonstrates how predictive modeling can be integrated into a real-time conversational assessment platform. LLMs were able to generate risk predictions and explanations based on the collected information. Such capabilities extend beyond the output of conventional ML classifiers and support user engagement, transparency, and better communication of risk.

Overall, this work shows that LLMs provide an effective and adaptable foundation for

digital health assessment tools. Their ability to generate explanations, adapt with minimal labeled data, and operate within interactive systems suggests a promising path forward for accessible and explainable decision support during public-health crises.

CHAPTER 7 FUTURE WORK

Several future directions arise naturally from the findings of this thesis and the areas noted in the conclusion of the initial study.

A primary direction is the expansion to larger and more diverse datasets. Because the current dataset was collected via a mobile application, it may underrepresent certain demographic or clinical groups. Incorporating data from clinical records, telemedicine transcripts, or more heterogeneous populations would improve robustness, reduce bias, and strengthen generalizability. Previous research also notes the importance of validating model performance in different populations and evolving epidemic contexts, especially as COVID-19 symptoms and variants change over time.

Another direction involves richer serialization strategies. This study compared list- and text-based formats, but additional formats: structured JSON-like representations, hybrid templates, or templates that emphasize temporal symptom patterns—may further improve consistency and interpretability. Future systems may also integrate multi-turn reasoning to refine predictions over multiple rounds.

Interpretability is another area of development. While attention maps and generated rationales offer intuitive insights, more rigorous interpretability frameworks could strengthen clinician trust. Methods such as attention-based attribution or prompting the model for structured explanations may help reveal how individual features contribute to risk predictions.

Several deployment-oriented opportunities also exist. Evaluating system behavior under real-world usage conditions, including latency, error recovery, and ambiguous user

responses—will be essential before clinical or public-health deployment. Mechanisms that detect uncertain predictions, escalate high-risk responses, or provide safety disclaimers would improve the system's reliability. As noted previously, mobile-based digital tools can play a meaningful role during outbreaks, but ensuring safety and clarity is critical.

Finally, the framework may be extended to other diseases and assessment scenarios. The approach of serializing structured features, fine-tuning LLMs in a few-shot setting, and integrating predictions with conversational explanations is general and not limited to COVID-19. Risk assessment for influenza, chronic conditions, or triage-oriented symptom evaluation are all viable extensions, provided appropriate datasets and validation procedures are available.

In summary, future work should focus on expanding datasets, enhancing serialization and interpretability, validating system behavior in broader contexts, and generalizing the methodology to other areas of digital health. These directions build directly on these findings while supporting stronger, safer, and more versatile health-assessment applications.

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APPENDIX

For the role of AIGC in this thesis, I only use ChatGPT for grammar checking, proof-reading, and revising the content I have already written. I did not use any AIGC tools to generate creative content.

ABSTRACT

GENERATIVE LARGE LANGUAGE MODEL-POWERED CONVERSATIONAL AI APP FOR PERSONALIZED RISK ASSESSMENT: CASE STUDY IN COVID-19

by

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December 2025

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Accurate prediction of COVID-19 severity risk remains an essential challenge for supporting timely intervention, resource planning, and personalized patient guidance. Traditional machine learning models have been useful in structured clinical prediction tasks, but they depend on curated datasets and do not provide explanations or interactive capabilities for users. Recent advances in large language models open new possibilities for combining predictive modeling with real-time conversational assessment.

This thesis investigates the use of generative large language models for COVID-19 severity prediction using structured symptom, demographic, and comorbidity information collected through a mobile self-assessment system. The thesis examines how prompt serialization formats, few-shot fine-tuning strategies, and model architectures influence predictive performance. The experimental framework evaluates several language models, including Flan-T5 variants, T0 models, and Llama 2, and compares them with traditional classifiers such as logistic regression, random forest, and XGBoost. The study incorporates probability-based classification, attention-driven interpretability, and an end-to-end

conversational inference pipeline.

The results show that fine-tuned language models consistently perform at the level of or better than traditional machine learning baselines, even when only a small number of examples are available. List-based serialization provides stable improvements in settings with very few examples, while text-based serialization improves as more demonstrations are introduced. The fine-tuned models also provide natural language explanations and highlight clinically meaningful feature patterns through attention mechanisms. These capabilities enable forms of clarity and transparency that standard classifiers cannot provide. In a conversational application, the system is able to gather missing information, guide users through the assessment, and communicate personalized risk estimates.

The thesis concludes that generative language models offer a practical and data-efficient approach for digital health risk assessment. They combine prediction, explanation, and interactive communication within a single modeling framework. The thesis also discusses limitations related to dataset size, generalizability across populations, and interpretability, and presents several directions for future work, including larger datasets, richer forms of serialization, improved attribution methods, and expanded evaluation in real-world digital health environments.

AUTOBIOGRAPHICAL STATEMENT

Mohammad Amin Roshani is a Ph.D. student in Computer Science at Wayne State University, working under the supervision of Dr. Dongxiao Zhu. He received his Bachelor of Computer Science from the University of Tehran. His research focuses on applying artificial intelligence to medicine and bioinformatics. He develops large language models and AI agent systems for clinical and biomedical tasks, emphasizing trustworthiness, interpretability, and real-world deployment. His current work involves extracting Social Determinants of Health (SDOH) from the MIMIC dataset and leading a newly funded collaboration with the School of Nursing to build an AI-assisted health application for older adults that streamlines data sharing, mobility tracking, and clinical communication. Previously, he contributed to an NIH-funded project on COVID-19 severity risk assessment, with findings published in JMIR AI. His goal is to create reliable and human-centered AI systems that enhance decision-making and promote equitable, transparent healthcare.